

DATASET APPROVAL PROPOSAL

Hotel Reservation Analysis and Prediction

1. Dataset

I propose using the **Hotel Reservations Dataset** published on Kaggle. It contains **36,275 reservation records and 19 columns**, with a useful mix of numerical and categorical variables such as guest numbers, length of stay, meal plan, room type, lead time, arrival date, market segment, previous booking behaviour, room price, special requests, and booking status. The dataset is public and shareable under the **CC BY 4.0 licence** and contains no directly identifying personal information.

Source: [Kaggle – Hotel Reservations Dataset](#)

2. Classification question

Can the available reservation details be used to predict whether a hotel booking will be cancelled? The categorical target will be the existing **booking_status** column, with the classes *Canceled* and *Not_Canceled*.

3. Regression question

Can the characteristics of a reservation be used to predict its average room price? The numeric target will be the existing **avg_price_per_room** column. Potential predictors include room type, arrival month, number of guests, length of stay, meal plan, market segment, lead time, parking requirement, and special requests.

4. Why the project is useful

The dataset supports two practical hotel-management questions: anticipating cancellations and understanding room-price drivers. The same data can also support exploratory analysis, interactive filtering, scenario comparison, and a dashboard that explains the model predictions in clear business terms.